

AI Search Framework: Experiment 2 (E2) Empirical Evaluation Report

Query-Dependent SLM Fine-Tuning vs. 4-Tier Non-Fine-Tuned Baseline Ladder

Repository: dixitabhi1/SLM_PROJECT (Branch: exp/mentor-protocol-e1)

Benchmark Cohort: 8 Canonical Two-Domain Queries (100% Taxonomy Coverage)

Evaluation Methodology: Symmetrical Double-Blind Judging (64 trials, 16/tier)

Fairness Status: Pre-Flight Verified (11.85B < P_{Baseline} across all 4 tiers)

1. Executive Summary & Experimental Objectives

Experiment 2 (E2) investigates the empirical impact of Query-Dependent Specialist Fine-Tuning under the Mentor Experiment Protocol. In contrast to Experiment 1 (E1) where all SLM pool models were un-adapted, E2 selectively routes subtasks based on query domain requirements: technical coding/systems subtasks are routed to `phi3.5-ft-coding:latest` (3.82B parameters, local RTX 3050 GPU), while general and analytical formulation tasks are executed by the base general specialist (`meta-llama/LLama-3.1-8B-Instruct`). All 4 Baselines remain strictly un-adapted and non-fine-tuned, matched directly against cached E1 baseline responses for 100% paired-trial fidelity.

2. Pre-Flight Governance Assertions

- Hard Rule 13 (Distinct Models):** All systems map to distinct checkpoints on separate endpoints (0 collisions).
- Hard Rule 16(b) (Fairness Constraint Pre-Flight):** Combined participating SLM parameters (3.82B + 8.03B = **11.85B**) strictly < 20.0B < 32.0B < 72.7B < 120.0B across all tiers.

3. Overleaf Master Results Table — Experiment 2 (E2)

Baseline Tier	Baseline Model	Params	Framework Mode	SLM Wins	Draws	LLM Wins	Effective SLM Win (Q _S ≥ Q _L)	Quality Proximity [95% CI]	SLM Score	LLM Score	Mean ΔQ [95% CI]	Matched Gain vs E1
Tier 1 (~20B)	openai/gpt-oss-20b	20.0B	1-10 Holistic	1 (6.25%)	0	15	1 (6.25%)	0.4305 [0.3500, 0.5111]	2.81	7.56	-4.75 [-6.03, -3.47]	+0.062
			1-5 Criteria	1 (6.25%)	0	15	1 (6.25%)	42.71% [33.55%, 51.86%]	2.15	4.27	-2.12 [-2.72, -1.53]	+0.021
Tier 2 (~32B)	gemini-2.5-flash	32.0B	1-10 Holistic	0 (0.0%)	0	16	0 (0.0%)	0.4861 [0.4148, 0.5574]	2.81	7.44	-4.62 [-5.27, -3.98]	+0.000
			1-5 Criteria	0 (0.0%)	0	16	0 (0.0%)	45.31% [37.37%, 53.25%]	2.12	4.31	-2.19 [-2.50, -1.87]	-0.021
Tier 3 (~72B)	Qwen/Qwen2.5-72B-Instruct	72.7B	1-10 Holistic	0 (0.0%)	0	16	0 (0.0%)	0.4514 [0.3849, 0.5179]	2.56	7.50	-4.94 [-5.54, -4.34]	-0.188
			1-5 Criteria	0 (0.0%)	0	16	0 (0.0%)	43.23% [35.25%, 51.21%]	2.00	4.27	-2.27 [-2.59, -1.95]	-0.021
Tier 4 (~120B)	openai/gpt-oss-120b	120.0B	1-10 Holistic	0 (0.0%)	0	16	0 (0.0%)	0.3194 [0.2231, 0.4158]	2.44	8.56	-6.12 [-6.99, -5.26]	+0.000
			1-5 Criteria	0 (0.0%)	0	16	0 (0.0%)	30.21% [20.36%, 40.05%]	1.96	4.75	-2.79 [-3.19, -2.40]	+0.042

4. Matched Gain Analysis (E1 vs. E2)

Baseline Tier	E1 Holistic QP	E2 Holistic QP	ΔQP Gain	E1 Mean ΔQ	E2 Mean ΔQ	Matched ΔQ Gain
Tier 1 (~20B)	0.4236	0.4305	+0.0069	-4.81	-4.75	+0.062
Tier 2 (~32B)	0.4861	0.4861	+0.0000	-4.62	-4.62	+0.000
Tier 3 (~72B)	0.4722	0.4514	-0.0208	-4.75	-4.94	-0.188
Tier 4 (~120B)	0.3194	0.3194	+0.0000	-6.12	-6.12	+0.000

6. Threats to Validity: Vendor-Lineage Disclosure & Cross-Model Validation

Vendor-Lineage Disclosure: Baseline Tier 2 (`gemini-2.5-flash` , 32.0B) and the primary evaluator (`gemini-3.1-flash-lite`) share vendor lineage (Google DeepMind). To empirically test and eliminate the potential threat of vendor self-preference bias, an independent cross-validation pass was executed using `qwen/qwen3.8-27b` on Groq API across all 16 symmetrical trials.

- **Verdict Agreement Rate: 93.75%** (15/16 trials) concordant with Gemini on winner determination.
- **Positional Swap Consistency: 87.5%** consistency under forward vs swapped presentation on Qwen 27B.
- **Score Alignment:** Qwen Holistic Mean: SLM 1.88, LLM 6.00 (Delta: -4.12) vs Gemini Holistic Mean: SLM 2.81, LLM 7.44 (Delta: -4.62).
- **Qualitative Defect Alignment:** Qwen independently identified the exact same technical defects as Gemini (e.g., wall convective coefficients using fluid Reynolds correlations in `V3_TD_01` , fixed coordinate Hamiltonian eigenvectors in `V3_TD_41` , fragmented C code in `V3_TD_51`), proving the outcome is an objective capability gap rather than evaluator self-preference.